

A COUNTEREXAMPLE TO BOX-HALF-INTEGRALITY OF THE INTERSECTION OF CROSSING SUBMODULAR FLOW SYSTEMS

MICHEL X. GOEMANS AND YUCHONG PAN

ABSTRACT. Abdi, Cornuéjols and Zambelli [Combinatorica 2024] proved that the intersection of two crossing submodular flow systems on a weakly connected digraph is totally dual integral, though not box-integral. Abdi subsequently conjectured that this polyhedral system is box-half-integral. We exhibit a counterexample to this conjecture.

We say that a family $\mathcal{C}$ of subsets of a ground set V is *crossing* if for all $U, W \in \mathcal{C}$ with $U \cap W \neq \emptyset$ and $U \cup W \neq V$, we have $U \cap W, U \cup W \in \mathcal{C}$, and that $\mathcal{C}$ is *cross-free* if for all $U, W \in \mathcal{C}$, we have $U \subseteq W$ or $W \subseteq U$ or $U \cap W = \emptyset$ or $U \cup W = V$. We say that a set function $f : \mathcal{C} \rightarrow \mathbb{R}$ is *crossing submodular* if for all $U, W \in \mathcal{C}$ with $U \cap W \neq \emptyset$ and $U \cup W \neq V$,

$$f(U \cap W) + f(U \cup W) \leq f(U) + f(W).$$

In [1], Abdi, Cornuéjols and Zambelli proved the following theorem:

Theorem 1 (Abdi, Cornuéjols and Zambelli, 2023, [1]). *Let $D = (V, A)$ be a weakly connected digraph. Let $\mathcal{C}_i$ be a crossing family over V and $f_i : \mathcal{C}_i \rightarrow \mathbb{Z}$ a crossing submodular function for $i = 1, 2$. Then*

$$\begin{aligned} y(\delta^+(U)) - y(\delta^-(U)) &\leq f_1(U), & \forall U \in \mathcal{C}_1, \\ y(\delta^+(U)) - y(\delta^-(U)) &\leq f_2(U), & \forall U \in \mathcal{C}_2, \end{aligned}$$

is totally dual integral.

In addition, they gave an instance for which this system is not box-integral. In a talk given in the “Combinatorics and Optimization” workshop at ICERM in 2023, Abdi proposed the following conjecture:¹

Conjecture 2. *Let $D = (V, A)$ be a weakly connected digraph. Let $\mathcal{C}_i$ be a crossing family over V and $f_i : \mathcal{C}_i \rightarrow \mathbb{Z}$ a crossing submodular function for $i = 1, 2$. Then*

$$\begin{aligned} y(\delta^+(U)) - y(\delta^-(U)) &\leq f_1(U), & \forall U \in \mathcal{C}_1, \\ y(\delta^+(U)) - y(\delta^-(U)) &\leq f_2(U), & \forall U \in \mathcal{C}_2, \end{aligned}$$

is box-half-integral.

¹See the last slide of their talk, available at https://app.icerm.brown.edu/assets/403/4951/4951_3700_Abdi_032820231400_Slides.pdf.

In this short note, we disprove this conjecture, making Theorem 1 more interesting.

Theorem 3. *There exist a weakly connected digraph $D = (V, A)$, crossing submodular functions $f_i : \mathcal{C}_i \rightarrow \mathbb{Z}$ for $i = 1, 2$ and $\ell, u \in \mathbb{Z}^A$ such that*

$$\begin{aligned} (1a) \quad & y(\delta^+(U)) - y(\delta^-(U)) \leq f_1(U), & \forall U \in \mathcal{C}_1, \\ (1b) \quad & y(\delta^+(U)) - y(\delta^-(U)) \leq f_2(U), & \forall U \in \mathcal{C}_2, \\ (1c) \quad & \ell_e \leq y_e \leq u_e, & \forall e \in A, \end{aligned}$$

is not half-integral.

Proof. Let $D = (V, A)$ be the weakly connected digraph in Figure 1a. Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be families of subsets of V depicted in Figures 1b and 1c, with corresponding set functions $f_i : \mathcal{C}_i \rightarrow \mathbb{Z}$ for $i = 1, 2$. Since each of $\mathcal{C}_1$ and $\mathcal{C}_2$ is a family of disjoint sets, $\mathcal{C}_1$ and $\mathcal{C}_2$ are trivially crossing families, and f_1 and f_2 are trivially crossing submodular. Let $\ell = -2 \cdot \mathbf{1} \in \mathbb{Z}^A$ and $u = \mathbf{1} \in \mathbb{Z}^A$. Let $y^* \in \mathbb{Z}^A$ be the vector defined in Figure 1d. It is easy to check that y^* is a feasible solution to the system (1).

To see that y^* is an extreme point of the polyhedron determined by the system (1), it suffices to exhibit 6 linearly independent inequalities that are tight at y^* . Since $y_e = u_e = 1$ for $e \in \{(v_1, v_2), (v_3, v_4), (v_5, v_6)\}$, we obtain 3 inequalities that are tight at y^* . In addition, it is easy to check $y(\delta^+(U)) - y(\delta^-(U)) = f_1(U) = 1$ for all $U \in \mathcal{C}_1$ and $y(\delta^+(U)) - y(\delta^-(U)) = f_2(U) = 0$ for all $U \in \mathcal{C}_2$, yielding 3 more inequalities that are tight at y^* . Moreover, it can be checked that these 6 inequalities are linearly independent (e.g., by computing the determinant of the matrix whose columns are these vectors). This gives 6 linearly independent inequalities that are tight at y^* , completing the proof. $\square$

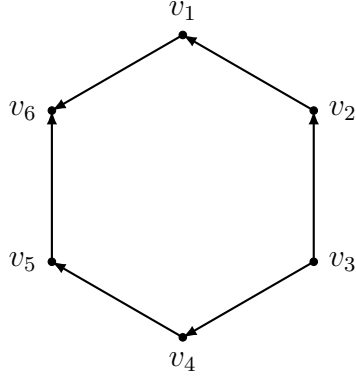
Remark. The intuition of the proof is the following. Let M_1 and M_2 be matrices with the same number of rows such that each row consists of at most one 1-entry and at most one -1 -entry, with all other entries equal to 0. Then M_1 and M_2 are submatrices of incidence matrices of digraphs and are hence totally unimodular. Let

$$M = \begin{bmatrix} M_1 & M_2 \end{bmatrix}.$$

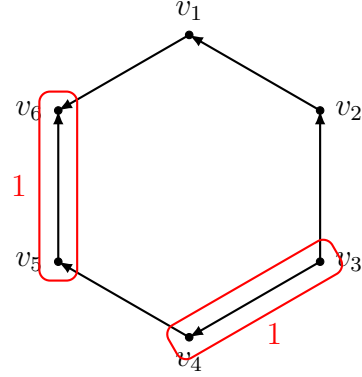
Note that M is not necessarily totally unimodular, and there exist instances such that $|\det(M)|$ can be made arbitrarily large. Take such a square matrix M with $|\det(M)| \geq 3$. Let b be an integral vector such that $(M^T)^{-1}b$ is not half-integral. In the counterexample given in the proof,

$$M = \left[\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & -1 & -1 \\ 1 & -1 & 1 \end{array} \right], \quad b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix},$$

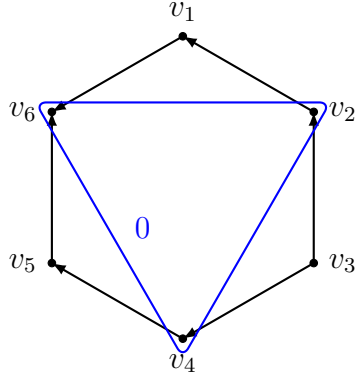
where $\det(M) = -3$.



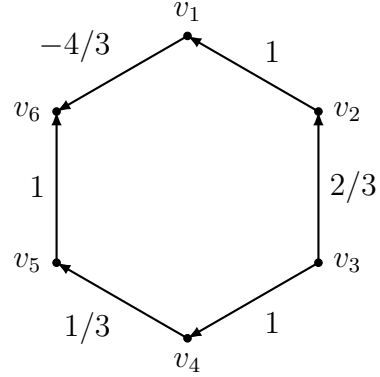
(A) The digraph $D = (V, A)$ in the proof of Theorem 3.



(B) The crossing family $\mathcal{C}_1$ and the crossing submodular set function $f_1 : \mathcal{C}_1 \rightarrow \mathbb{Z}$ in the proof of Theorem 3.



(C) The crossing family $\mathcal{C}_2$ and the crossing submodular set function $f_2 : \mathcal{C}_2 \rightarrow \mathbb{Z}$ in the proof of Theorem 3.



(D) A solution $y^* \in \mathbb{Z}^A$ to the system of linear inequalities in the proof of Theorem 3.

Let k_1 and k_2 be the numbers of columns of M from M_1 and from M_2 , respectively. Let m be the number of rows of M . Let $D_0 = (V, A_0)$ be a digraph, where $V = \{0, \dots, k_1\} \times \{0, \dots, k_2\}$ and A_0 has m arcs, one corresponding to each row of M . For each row of M , let α, α' be the indices of the 1-entries from M_1 and from M_2 (or 0 if not present), respectively, and β, β' the indices of -1 -entries from M_1 and from M_2 (or 0 if not present), respectively. Then this row corresponds to the arc $((\alpha, \alpha'), (\beta, \beta'))$.

Let $S_i = \{i\} \times \{0, \dots, k_2\}$ for each $i \in [k_1]$. Let $T_i = \{0, \dots, k_1\} \times \{i\}$ for each $i \in [k_2]$. Let $\mathcal{C}_1 = \{S_1, \dots, S_{k_1}\}$ and $\mathcal{C}_2 = \{T_1, \dots, T_{k_2}\}$. Let $g_1 : \mathcal{C}_1 \rightarrow \mathbb{Z}$ be defined by $g_1(S_i) = b_i$ for all $i \in [k_1]$, and let $g_2 : \mathcal{C}_2 \rightarrow \mathbb{Z}$ be defined by $g_2(T_i) = b_{i+k_1}$ for all $i \in [k_2]$. For $i = 1, 2$, since the sets in $\mathcal{C}_i$ are disjoint, $\mathcal{C}_i$ is trivially a crossing family and g_i is trivially crossing submodular. The

system $M^T y \leq b$ is exactly

$$\begin{aligned} y(\delta^+(U)) - y(\delta^-(U)) &\leq g_1(U), & \forall U \in \mathcal{C}_1, \\ y(\delta^+(U)) - y(\delta^-(U)) &\leq g_2(U), & \forall U \in \mathcal{C}_2. \end{aligned}$$

Let $y_0 \in \mathbb{Q}^A$ be the unique solution to the system $M^T y = b$.

Now, we arbitrarily add edges to D_0 to create a weakly connected digraph $D = (V, A)$. For each new arc $e \in A \setminus A_0$, add a constraint $y_e \leq 1$. This results in a system $(M')^T y \leq b'$, where

$$M' = \begin{bmatrix} M & M'' \\ 0 & I \end{bmatrix}, \quad b' = \begin{bmatrix} b + b'' \\ \mathbf{1} \end{bmatrix},$$

where each row of M'' is the vector $\gamma(U) := \delta_{A \setminus A_0}^+(U) - \delta_{A \setminus A_0}^-(U)$ for $U \in \mathcal{C}_1$ and for $U \in \mathcal{C}_2$, and where each component of b'' is $\mathbf{1}^T \gamma(U)$ for $U \in \mathcal{C}_1$ and for $U \in \mathcal{C}_2$. By Schur's formula, $\det(M') = \det(M) \cdot \det(I) = \det(M)$. Hence, $|\det(M')| \geq 3$. Let

$$y^* = \begin{bmatrix} y_0 \\ \mathbf{1} \end{bmatrix}.$$

It is easy to check that y^* is the unique solution to the system $(M')^T y = b$.

With the procedure described above, one can find counterexamples to Conjecture 2 with arbitrarily large sizes.

REFERENCES

- [1] Ahmad Abdi, Gérard Cornuéjols, and Giacomo Zambelli. Arc connectivity and submodular flows in digraphs. *Combinatorica*, 44(5):1069–1090, 2024.